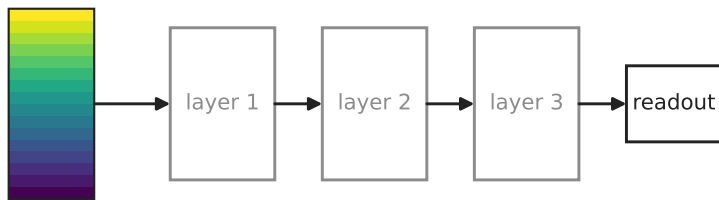


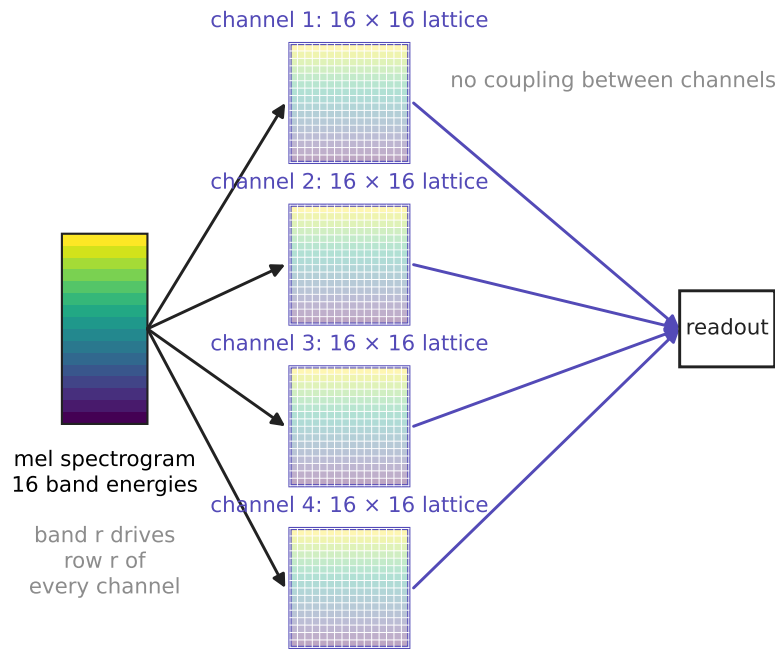
(a) Layers: stacked, one after another (depth)



mel spectrogram
16 band energies

each layer transforms the previous layer's output;
a deep network or a transformer stacks many

(b) Channels: side by side, all driven by the same input (width)



mel spectrogram
16 band energies

band r drives
row r of
every channel

each channel has its own coupling kernel and natural frequencies; within a channel every oscillator is coupled to every other. The readout reads all channels' signals together, as the attention heads of one transformer layer are read together.